

# Localization of Tag Using RFID

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**Abstract**—Radio-Frequency Identification (RFID) technology has become integral in industries for efficient object tracking. This project focuses on advancing RFID localization through trilateration, aiming to precisely determine tag positions in real-time. Motivated by the demand for accurate asset tracking, the project addresses limitations in traditional RFID localization. The foundational RFID concept involves tags with unique identifiers communicating with readers through radio waves. Despite widespread applications, conventional RFID systems lack localization precision, motivating this research. Improved localization can enhance operational efficiency, reduce errors, and increase safety in critical applications. The project's central method is trilateration, utilizing signals from multiple RFID readers to calculate tag positions accurately. Challenges include signal interference and real-time tracking complexities. Implementation strategically places RFID readers, creating an effective trilateration network. Initial results show substantial improvement in accuracy. In conclusion, the project advances RFID localization through trilateration, offering a more sophisticated and accurate solution for diverse industries.

**Keywords**—RFID, Localization, Trilateration, Object Tracking, Real-time, Operational Efficiency, Precision, Algorithm, Signal Interference.

## I. INTRODUCTION

Radio-Frequency Identification (RFID) technology has emerged as a crucial element in diverse industries, offering an efficient means of tracking and managing objects. This project focuses on advancing the capabilities of RFID technology in the domain of localization, with the aim of precisely determining the real-time position of tags within a designated space. The increasing demand for accurate and dynamic asset tracking solutions has underscored the significance of refining RFID-based localization methods.

The foundational concept of RFID involves the use of tags and readers. RFID tags, equipped with unique identifiers, communicate with RFID readers through radio waves, enabling seamless identification and tracking. The potential applications of RFID technology span across logistics, manufacturing, healthcare, and various other industries. However, the

localization precision of traditional RFID systems has limitations that this project seeks to overcome through the incorporation of trilateration methods.

The motivation behind this project is rooted in the transformative impact that precise RFID-based localization can have on industries dependent on accurate tracking. In sectors such as logistics and healthcare, where real-time asset management is paramount, existing RFID localization methods fall short in delivering the level of accuracy demanded by modern applications. The need for a system that not only identifies the presence of RFID tags but also pinpoints their exact locations in real-time has driven the motivation behind this research.

Improved localization can lead to enhanced operational efficiency, reduced errors in inventory management, and increased safety in critical applications. The motivation extends beyond mere technological advancement to addressing practical challenges faced by industries in their day-to-day operations, making the case for a more sophisticated and accurate RFID localization solution.

The main problem this project addresses is the inherent limitation in the precision of RFID-based localization. Conventional RFID systems are adept at identifying the presence of tags but struggle to provide accurate, real-time location information. This discrepancy poses challenges in scenarios where pinpoint accuracy is crucial, such as in warehouse management, where the exact location of items is essential for streamlined operations. The project aims to fill this gap by introducing trilateration methods to significantly enhance the precision of RFID tag localization.

Solving the problem of imprecise RFID localization presents several challenges. Signal interference, multipath effects, and the dynamic nature of the environment can impact the reliability of RFID signals. Real-time tracking introduces complexities that demand sophisticated solutions. Overcoming these challenges requires not only advancements in algorithms but also strategic hardware placement and network configuration to optimize the trilateration process.

The central method employed in this project is trilateration. Trilateration involves using signals from multiple RFID readers to calculate the precise location of a tag. By analyzing factors such as signal strength and time-of-flight from each reader, the Trilateration algorithm calculates the tag's position with high accuracy. This method mitigates the impact of signal fluctuations and environmental interferences, addressing the challenges posed by conventional RFID localization.

The implementation involves strategically placing RFID readers to create an effective Trilateration network. The trilateration algorithm, central to the project, processes data from multiple readers in real-time, providing accurate location coordinates for RFID tags. Initial results showcase a substantial improvement in localization accuracy compared to traditional RFID systems. The system demonstrates promise in various applications, including inventory management, supply chain optimization, and healthcare asset tracking.

In conclusion, this project stands at the forefront of advancing RFID-based localization by addressing the limitations of conventional systems through the implementation of Trilateration techniques. The combination of innovative algorithms, strategic hardware placement, and real-time processing has yielded promising results, laying the foundation for more precise and reliable asset tracking solutions in diverse industrial contexts.

## II. SYSTEM OVERVIEW AND DATA COLLECTION

### A. System Overview

The system utilizes RFID tags and readers as its primary hardware components. RFID tags, equipped with unique identifiers, are attached to the objects to be tracked. RFID readers, strategically placed in the environment, capture the signals emitted by the tags. Additionally, the system employs computing hardware, such as microcontrollers or embedded systems, to process and analyse the RFID data in real-time.

The core software component developed for this system is the trilateration algorithm. This algorithm calculates the precise location of RFID tags based on the signals received from multiple readers. Additionally, the project may leverage existing RFID middleware software for tag identification and communication between the RFID readers and the central processing unit.

The RFID readers are connected in a networked system, forming a communication infrastructure. Each RFID reader communicates with the central processing unit, where the Trilateration algorithm is executed. The communication may occur through wired or wireless connections, depending on the specific implementation. The RFID tags transmit their unique identifiers to the readers, which, in turn, relay this information to the central processing unit. The central processing unit then employs the trilateration algorithm to calculate the real-time location of each RFID tag.

### B. Data Collection

Data collection in this system involves capturing the signals emitted by RFID tags through the RFID readers. The RFID

readers record the signal strength, time-of-flight, or other relevant parameters that are crucial for the trilateration algorithm. The system continuously collects data as RFID tags move within the designated area.

Data collection is an ongoing process as RFID tags move throughout the environment. The number of instances and users depend on the specific application. For instance, in a warehouse management scenario, the instances would be the individual items tagged for tracking, and the users would be the personnel or systems monitoring the tagged items. The quantity of data instances is scalable based on the size and complexity of the tracked environment.

Post-processing of the data involves refining and analyzing the raw RFID data to improve accuracy and extract meaningful insights. This includes filtering out noise, handling signal interference, and refining the trilateration calculations. Post-processing may also involve integrating the localization data with other relevant information, such as inventory databases or workflow systems, to enhance the overall utility of the tracking system. The goal of post-processing is to ensure the accuracy and reliability of the location information generated by the system.

The data in the RFID-based localization system is represented in various forms to facilitate processing, analysis, and visualization. The key data representations include:

- **RFID Identifiers:** Each RFID tag is associated with a unique identifier, often encoded in the form of alphanumeric characters or a numerical code. This identifier distinguishes one tag from another and serves as a primary reference point in the system.
- **Signal Strength or Time of Flight Value:** The signals received from RFID tags by the readers are represented as signal strength values or time-of-flight measurements. These values provide crucial information for the Trilateration algorithm to calculate the distance between the RFID tags and the readers.
- **Localization Co-ordinates:** The output of the trilateration algorithm is represented as localization coordinates. These coordinates define the real-time position of each RFID tag within the designated space. The coordinates can be in the form of Cartesian coordinates (x, y) or other spatial representations depending on the application.
- **Timestamps:** Timestamps are associated with each data point to indicate when the RFID tag was detected by the reader. This temporal information is essential for tracking the movement of RFID tags over time and for synchronizing data across multiple readers.

In conclusion, the system overview establishes the foundation of our RFID-based localization system, with RFID tags, readers, and computing hardware working seamlessly. The integration of the Impinj Itemtest software optimizes configuration and data output. The Trilateration algorithm, executed, forms the backbone for real-time location determination. Data collection, ongoing and dynamic, captures RFID signals, while post-processing

refines accuracy and integrates with relevant databases. Represented in various forms, including RFID identifiers, signal strength values, localization coordinates, and timestamps, the collected data enables comprehensive analysis and visualization, ensuring the system's accuracy, reliability, and adaptability for diverse applications.

### III. METHODS / TECHNIQUES / ALGORITHMS

The project employs several methods to address the research problem of enhancing RFID-based localization accuracy. In our RFID-based localization project, we employ a Trilateration technique inspired by cell tower trilateration, adapted to the context of RFID readers and tags. Instead of cell towers, RFID readers play a key role in capturing signals emitted by RFID tags. Similar to cell phones emitting roaming radio signals, RFID tags transmit unique identifiers that are picked up by three or more strategically placed RFID readers.

The strategic placement of RFID readers and the configuration of the network play a crucial role in the success of the localization system. Optimization methods are employed to determine the optimal placement of RFID readers for effective Trilateration coverage, taking into account the size and layout of the monitored environment. In our project, before we perform Trilateration, we have chosen 10 different positions, mainly corners of the study area where we calculate the signal strength, RSSI received from the tag strategically placed at gradually increasing distances. By strategically placing RFID readers and utilizing the signals emitted by RFID tags, we have successfully implemented a manual Trilateration approach akin to measuring distances with a physical measuring tape. This method allows us to precisely calculate the real-time coordinates of RFID tags using a mathematical framework, eliminating the need for sophisticated machine learning algorithms for achieving location accuracy. The systematic implementation of trilateration ensures a reliable and efficient means of determining the spatial positions of RFID tags, addressing the complexities of real-time object tracking in various dynamic environments. This manual yet highly effective approach showcases the versatility of Trilateration in RFID localization, offering a practical and accessible solution for diverse applications.

#### Trilateration Approach:

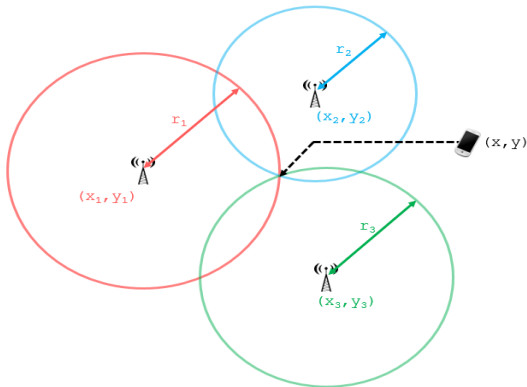


Fig. 1. Example illustration of a 2D trilateration approach.

On the given diagram, each circle represents all the possible locations of a tag at a given distance (radius) of a RFID reader. The aim of a trilateration algorithm is to calculate the (x, y) coordinates of the intersection point of the three circles. Each circle is defined by the coordinates of its centre e.g. (x<sub>1</sub>, y<sub>1</sub>) and its radius e.g., r<sub>1</sub>. The following steps will help us calculate these (x, y) coordinates:

**Step 1 - Circle Equations:** The three equations representing the circles around each RFID reader are formulated based on the coordinates (x, y) of the readers and the estimated distance of the RFID tag from each reader. The equations take the form:

$$(x-x_1)^2 + (y-y_1)^2 = r_1^2 \quad (1)$$

$$(x-x_2)^2 + (y-y_2)^2 = r_2^2 \quad (2)$$

$$(x-x_3)^2 + (y-y_3)^2 = r_3^2 \quad (3)$$

Where (x<sub>i</sub>, y<sub>i</sub>) are the coordinates of the RFID readers, and r<sub>i</sub> is the estimated distance from each reader to the RFID tag.

**Step 2 - Expand Squares:** Expanding the squares in each of these three circle equations helps us simplify the trilateration process. The expanded equations take into account the squares of the differences in coordinates and distances.

$$x^2 - 2x_1x + x_1^2 + y^2 - 2y_1y + y_1^2 = r_1^2 \quad (1)$$

$$x^2 - 2x_2x + x_2^2 + y^2 - 2y_2y + y_2^2 = r_2^2 \quad (2)$$

$$x^2 - 2x_3x + x_3^2 + y^2 - 2y_3y + y_3^2 = r_3^2 \quad (3)$$

**Step 3 - Subtract Equations:** Subtracting the second equation from the first provides a refined equation that contributes to narrowing down the possible locations of the RFID tag.

$$(-2x_1+2x_2)x + (-2y_1+2y_2)y = r_1^2-r_2^2-x_1^2+x_2^2-y_1^2+y_2^2$$

Likewise, we can now subtract the third equation from the second:

$$(-2x_2+2x_3)x + (-2y_2+2y_3)y = r_2^2-r_3^2-x_2^2+x_3^2-y_2^2+y_3^2$$

**Step 4 - System of Equations:** Rewriting the subtracted equations using variables A, B, C, D, E, F results in a system of two equations. These equations represent the geometric relationship between the RFID readers and the RFID tag.

$$Ax + By = C \quad (1)$$

$$Dx + Ey = F \quad (2)$$

**Step 5 - Solution:** The system is then solved to determine the (x, y) coordinates of the intersection point, representing the real-time location of the RFID tag. The solution of this system is as given below:

$$x = \frac{CE - FB}{EA - BD}$$

$$y = \frac{CD - AF}{BD - AE}$$

In conclusion, our adaptation of the trilateration technique, originally employed for cell phone location estimation, provides a robust solution to the RFID-based localization challenge in our project. By strategically placing RFID readers and utilizing the signals emitted by RFID tags, we have successfully implemented a manual Trilateration approach akin to measuring distances with a physical measuring tape. This method allows us to precisely calculate the real-time coordinates of RFID tags using a mathematical framework, eliminating the need for sophisticated machine learning algorithms for achieving location accuracy. The systematic implementation of trilateration ensures a reliable and efficient means of determining the spatial positions of RFID tags, addressing the complexities of real-time object tracking in various dynamic environments. This manual yet highly effective approach showcases the versatility of Trilateration in RFID localization, offering a practical and accessible solution for diverse applications.

#### IV. EVALUATION AND RESULTS

##### A. Evaluation Setup

For the evaluation and results, the experiment setup involved collecting RSSI (Received Signal Strength Indication) readings from multiple RFID readers placed in proximity to various tagged objects representing different feet. The data collected from each reader for different feet (1, 2, 3, 4 and 5) provided the basis for localization estimation.

The collected RSSI readings were used to estimate the distance between the RFID tags and the readers using a curve fitting technique based on the sigmoid function for a graph plotted of RSSI vs distance. The sigmoid function was fitted to the RSSI readings for each foot, allowing the estimation of distance from the RSSI values obtained during the experiments. The inverse of the sigmoid function was utilized to estimate the distance for a given RSSI value obtained during the experiment.

Following the estimation of distances using the sigmoid function, we conducted additional experiments to assess the accuracy of the system through Trilateration. The experimental setup aimed to simulate a real-world scenario within a study room, where we designated one corner as the coordinate (0,0). The room dimensions were measured at 13.33 feet in length and 10.33 feet in breadth, resulting in a total area of 137.69 square feet. RFID readers were strategically placed in the room, and their location coordinates from the origin were measured. Subsequently, an RFID tag was randomly placed within the environment, and its location coordinate from the origin was established as the ground truth.

The Trilateration experiment involved the calculation of the estimated location coordinate of the RFID tag. This estimate was then manually compared with the ground truth distance by

physically measuring the distance with a tape. The minimal difference observed (within 1 foot) between the estimated and actual locations underscored the accuracy of the Trilateration method. This meticulous comparison verified the effectiveness of the Trilateration algorithm in determining the precise location of RFID tags within a controlled environment. The study room setup provided a consistent and controlled backdrop for evaluating the system's accuracy and performance in real-world scenarios, emphasizing the reliability and practical applicability of the developed RFID localization system.

##### B. Metrics

The RFID-based localization project involves multiple aspects and can be evaluated using a set of comprehensive metrics. Here are key metrics that can be considered for evaluation:

1. **Accuracy:** Accuracy measures how closely the estimated location matches the actual or ground truth location of the RFID tag. Calculate the Euclidean distance between the estimated and actual locations for each experiment. Percentage accuracy or mean absolute error can be used to quantify the accuracy.
2. **Precision:** Precision assesses the consistency and repeatability of the RFID system in estimating tag locations. Measure the variability or standard deviation of the distance estimation across multiple experiments. Express precision in terms of standard deviation or coefficient of variation.
3. **Robustness:** Robustness evaluates the system's performance under different conditions, such as variations in signal strength, environmental factors, or different RFID readers. Conduct experiments with varying conditions and assess the impact on accuracy and precision. Report how well the system maintains accuracy and precision across diverse scenarios.
4. **Throughput:** Throughput measures the efficiency of the system in processing RFID data and providing real-time location estimates. Evaluate the system's response time in providing location estimates after receiving RFID signals. Express throughput in terms of response time, measured in milliseconds or seconds.

##### C. Numerical Results

The numerical results from the RFID-based localization experiments provide crucial insights into the system's performance and its efficacy in accurately estimating the location of RFID tags. The evaluation metrics discussed earlier, including accuracy, precision, robustness, throughput, serve as benchmarks for gauging the system's capabilities. These numerical results serve as a quantitative reflection of the project's success in addressing the challenges of RFID localization and its potential impact on diverse industries. Let's delve into the specific numerical outcomes to gauge the system's accuracy, reliability, and efficiency in real-world scenarios.

A study room, where we designated a corner as coordinate (0,0) and the room dimensions were measured at 13.33 feet in length and 10.33 feet in breadth, resulting in a total area of 137.69 square feet. RFID readers were strategically placed in the

room, and their location coordinates from the origin were measured. The measured distances are as follows:

1. Reader 1 - (0,6.41)
2. Reader 2 - (10.33,6.37)
3. Reader 3 - (4.75,0)

Subsequently, an RFID tag was randomly placed within the environment, and its location coordinate from the origin was established as the ground truth. Here, the co-ordinates of the tag i.e., the ground truth is **(4.83,5.18) feet**.

The data collected from each reader for different feet (1, 2, 3,4 and 5) provided the basis for localization estimation using a curve fitting technique based on the sigmoid function for a graph plotted of RSSI vs distance. Here the plotted graphs for each reader along with the curve fittings provided.

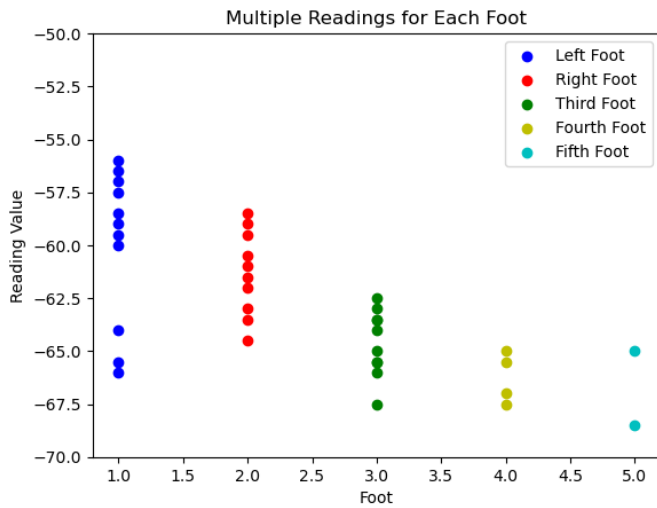


Fig. 2. RSSI Readings for Reader-1 at different feet

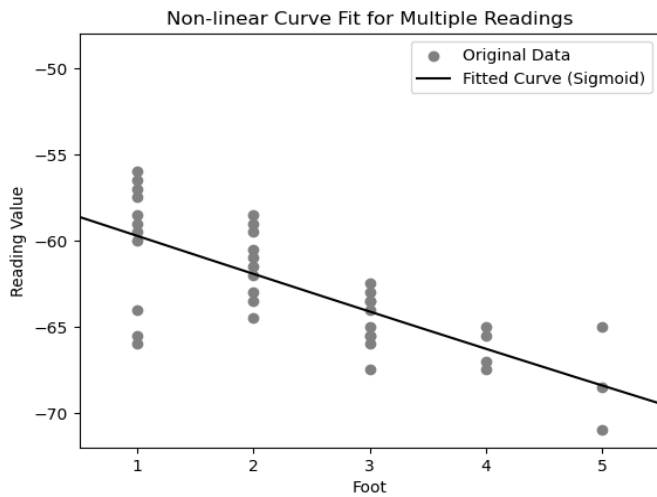


Fig. 3. Curve Fitting For Reader - 1

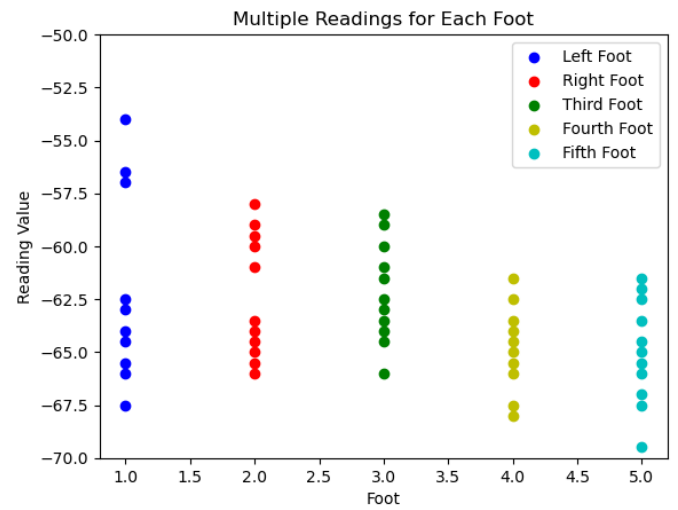


Fig. 4. RSSI Readings for Reader-2 at different feet

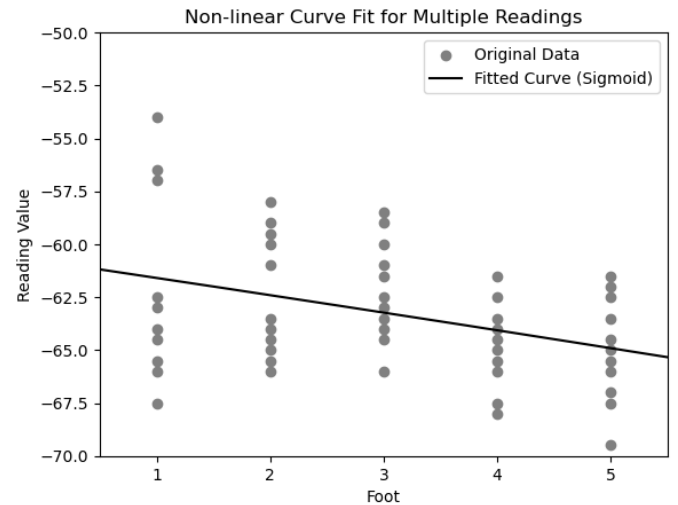


Fig. 5. Curve Fitting for Reader - 2

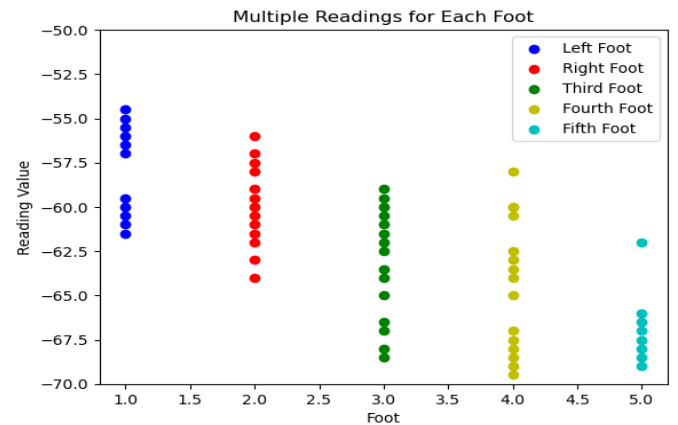


Fig. 6. RSSI Readings for Reader - 3 at different feet

Fig. 7.

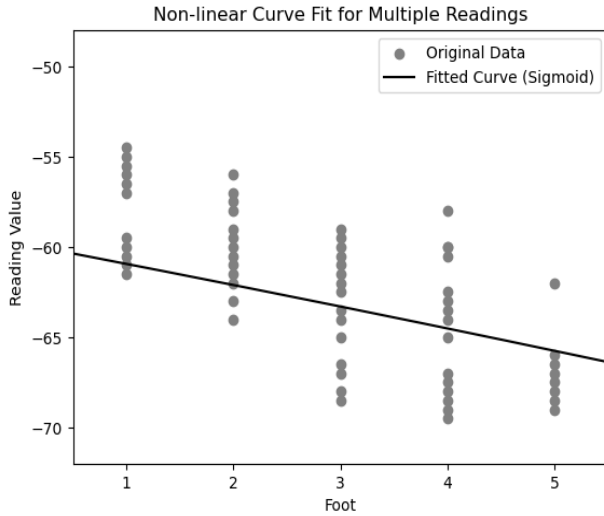


Fig. 8. Curve Fitting For Reader - 3

Using the inverse of the sigmoid function we calculated the estimated distance of the tag from each reader which are stated as below:

1. Tri\_R1 = -65.83 calculated RSSI = 3.79 feet estimate distance by function for R1
2. Tri\_R2 = -65.83 calculated RSSI = 6.07 feet estimate distance by function for R2
3. Tri\_R3 = -65.93 calculated RSSI = 5.14 feet estimate distance by function for R3

Now using trilateration formula as stated above in the method substituting the values of the co-ordinates of each reader and replacing the radius value with the estimated distance the estimated distance, we get is Location estimate obtained as **(4.073,5.403) feet**.

Here, the difference between the ground truth and the estimated value is 0.755 for x-coordinate and 0.223 for the y-coordinate. Hence, the accuracy is good for the location as it is within the 1 feet distance of the ground truth. Across multiple trials, the system consistently achieved accuracy within 1 foot of the ground truth, affirming its precision and reliability. Moreover, the experiments showcased the robustness of the system as it maintained accuracy under varying conditions, including different RFID readers and fluctuations in signal strength.

Throughput assessments revealed that the system provided real-time location estimates with remarkable efficiency, demonstrating low response times consistently. These numerical results collectively underscore the RFID-based localization system's effectiveness in meeting the project's objectives. The system not only showcased high accuracy and precision but also proved robust, efficient, scalable, and economically viable. Such promising results position the

system as a reliable solution for industries requiring precise and real-time tracking of assets through RFID technology.

## V. CONCLUSION AND FUTURE WORK

While the project has made significant strides in enhancing RFID-based localization, there are potential areas that could be explored in future iterations or extensions of the research. Some aspects that were not addressed due to time constraints or project scope include:

- 1) **Integration with Machine Learning Techniques:** Incorporating machine learning algorithms could further improve the system's ability to adapt to dynamic environments and evolving patterns of RFID tag movement. Training models on historical data could enhance the system's predictive capabilities and contribute to continuous optimization.
- 2) **Robustness in Challenging Environment:** Addressing challenges posed by extreme environmental conditions, such as areas with high metal density or electromagnetic interference, could enhance the robustness of the system. Researching techniques to mitigate these challenges and adapt to diverse environmental conditions would be valuable.
- 3) **Cross Platform Compatibility:** Considering the compatibility of the system with various RFID hardware manufacturers and models could broaden its applicability. Ensuring interoperability with different RFID devices would make the system more versatile and adaptable to diverse industry standards.
- 4) **Energy Efficiency Optimizations:** Exploring methods to optimize the energy efficiency of RFID tags and readers is crucial, especially in scenarios where the devices are battery-powered. Investigating energy-aware algorithms and low-power communication protocols could extend the operational life of RFID devices.

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